

PY4109 - Advanced Computational Physics

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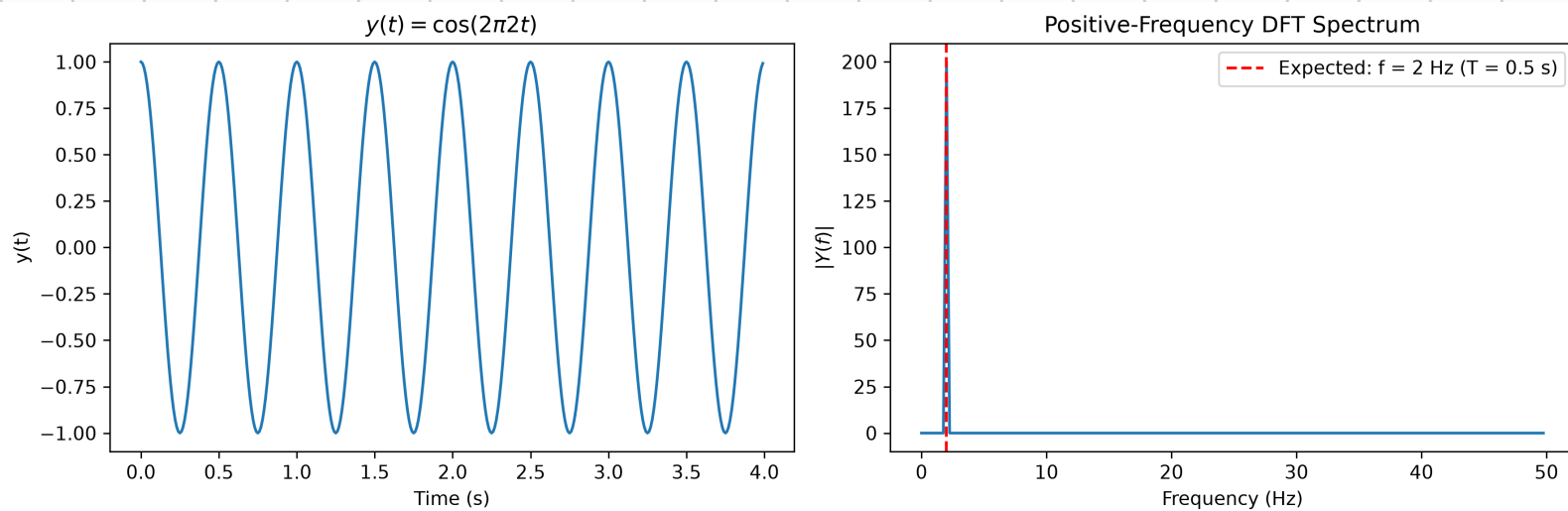
Declaration: I hereby declare all submitted work is my own and was created with the aid of lecture notes. Any external sources have been appropriately cited.

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Code has been submitted along with this file.
The code is commented explaining the workflow.
This pdf contains written components + compilation of all graphs generated

Problem 1 - The Fourier Transform

1. Write a piece of code which calculates the discrete Fourier transform. Verify your code works by calculating the Fourier Transform of $y(t) = \cos(2\pi 2t)$. You are free to define your own internals and time resolution for t . The output of this program should be a plot of $y(t)$ versus t and a plot of $Y(f)$ versus f . The plot of the Fourier spectrum should also have the known periods marked with vertical lines. (4 marks)



Defined intervals:

$\Delta t = 0.01$ uniformly sampled 0 - 4 seconds.

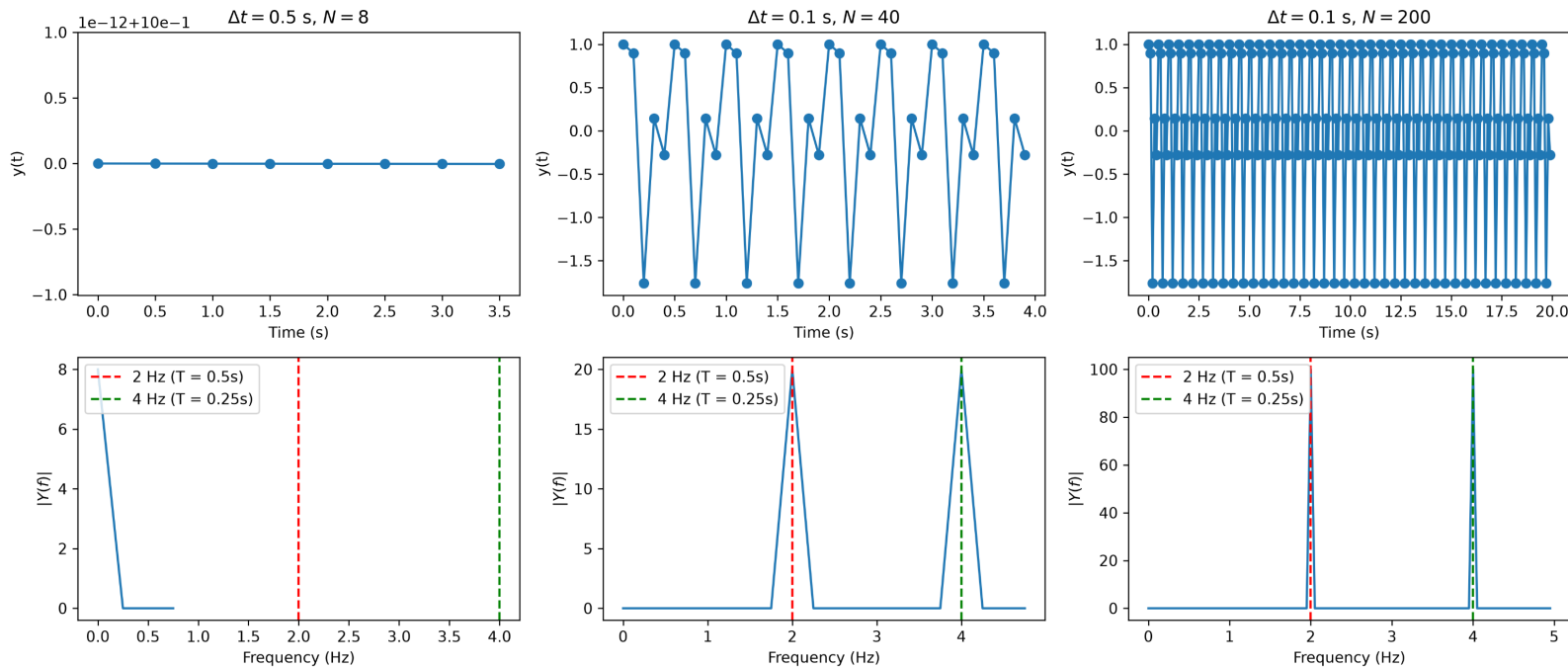
Expected frequency $f = 2$ Hz $T = 1/f = 0.5$ s

2. Compute the DFT for $y(t) = \cos(2\pi 2t) + \sin(2\pi 4t)$ for the following intervals (min,max,step):

1. $t = \text{range}(0,4,0.5)$
2. $t = \text{range}(0,4,0.1)$
3. $t = \text{range}(0,20,0.1)$

Using these results, comment on the accuracy of the recovered periods as both a function of the number of samples and the time step between the samples. (5 marks)

DFT of $y(t) = \cos(2\pi 2t) + \sin(2\pi 4t)$



Comment on Results:

- Signal has expected frequencies 2 Hz and 4 Hz
- The $t = \text{np.arange}(0,4,0.5)$ means arranging it so we take a measurement every 0.5s ($\Delta t = 0.5$)
- For $\Delta t = 0.5$ we are taking 8 measurements ($N = 4/0.5 = 8$)
- The DFT then produces 8 complex Fourier coefficients, which contain $2N$ independent real numbers (original DFT is real so DFT has conjugate symmetry)

$r = \text{arange}(N)$

We define frequency as $\rightarrow f = m/N\Delta t$

So the largest independent frequency index is $m = N/2$ because of the mirroring

$\leftarrow$ total duration is $N\Delta t$, so frequency spacing is $1/N\Delta t$

So then $f_{\max} = \frac{N/2}{N \Delta t} = \frac{1}{2 \Delta t}$ Nyquist Frequency

Applying this to 1st scenario

$$f_{\max} = \frac{8/2}{8(0.5)} = \frac{1}{2(0.5)} = 1 \text{ Hz}$$

Which is what we see, the first DFT plot stops at around 1 Hz

In fact the signal itself is basically a flat line. That's because we aren't sampling enough to get actual shape of the signal.

2nd scenario. $N = 4/0.1 = 40$

$$f_{\max} = \frac{40/2}{40/0.1} = \frac{1}{2(0.1)} = 5 \text{ Hz}$$

This allows for both expected frequencies of 2 Hz and 4 Hz

Third scenario keeps $f_{\max} = 5 \text{ Hz}$ but now we have 200 samples

$$N = 20/0.1 = 200$$

We are able to observe the signal for longer to get more accurate results

We are testing frequencies more closely spaced!

$$\Delta f = 1/N \Delta t$$

Run 2 $\Delta f = 1/40(0.1) = 0.25$

Run 3 $\Delta f = 1/200(0.1) = 0.05$

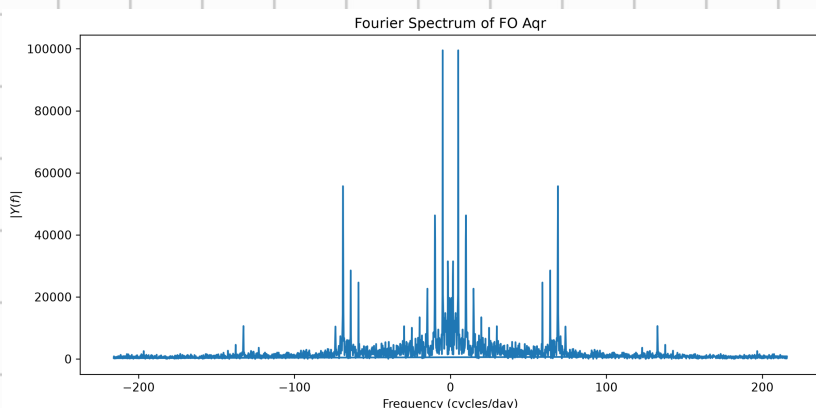
So if the expected frequency were something like 2.13Hz, the second run could only peak at 2 or 2.25, whereas the 3rd run has more finely spaced bins

In the graph you can see narrower and taller peaks. Because 2Hz falls within the bins for the second run, the difference in resolution can't be seen as much

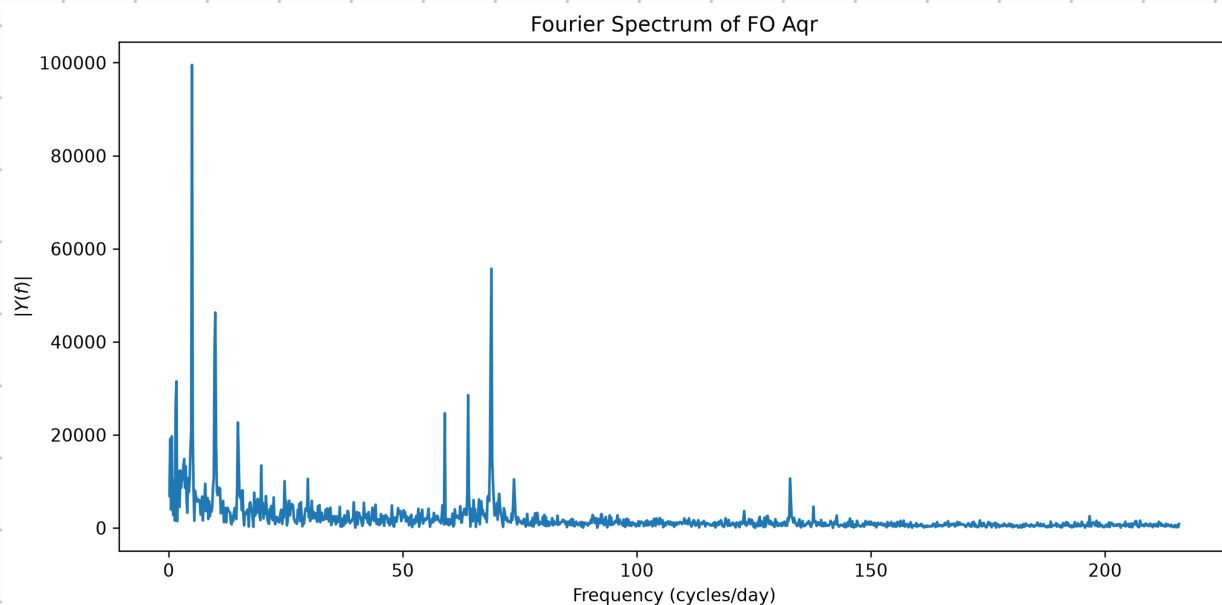
3. Distributed with this problem set is a time series of a binary star system. Verify whether the condition to apply a FT to these data are met. If it is not, write a piece of code which can manipulate the data to meet this condition. (2 marks)

In code file.

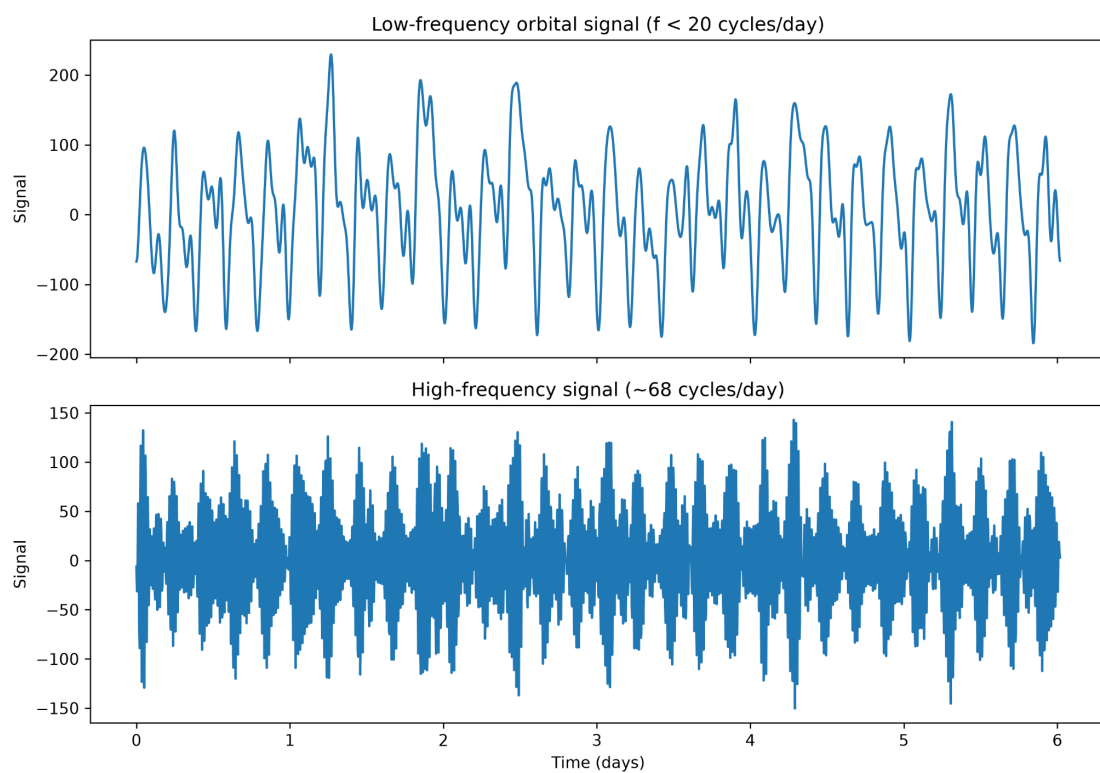
4. The Fourier Transform can be used as a high pass or low pass filter to isolate signals of interest. The time series discussed in the previous problem displays multiple periodicities. These can be broken down into two main components: a series of low frequency components ($f < 20$ per day) due to the orbital motion of the companion star in the binary, and some high frequency components ($f \sim 68$ cycles per day). By using appropriate frequencies filters, produce a plot of each of the main signals versus time. You can use a FFT package if necessary (`np.fft` in python or `fftw3.h` in C++). (5 marks)



→ extra graph where you can see the mirroring.



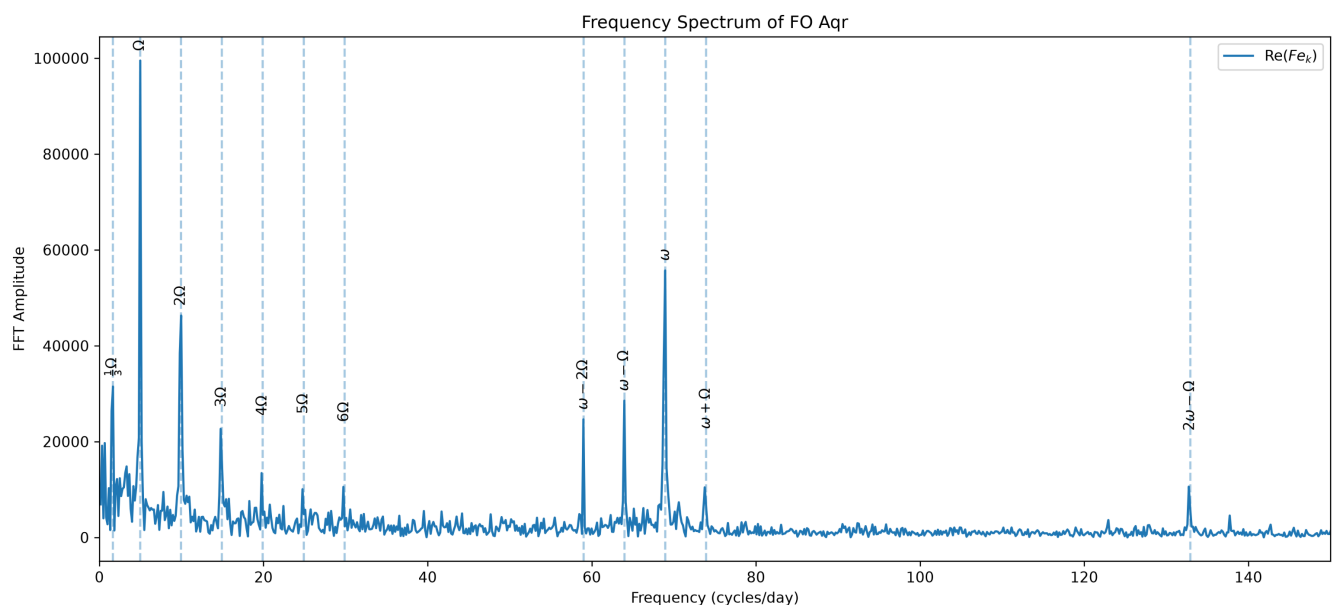
only plotting $f > 0$



Low frequency filter retains frequencies below 20 cycles/day, showing slower variations in the light curve.

High frequency isolates signal between 50-75 cycles/day, producing much more rapid oscillations

5. The two main periods in the frequency spectrum of the previous example are at a frequency of ~ 5 cycles per day (which we will label as Ω) and ~ 69 cycles per day (which we will label as ω). Identify the relationships between all other significant peaks and these two periods in the spectrum and submit a plot with all of these peaks clearly labelled (an example is given below with Ω and ω labelled). (2 marks)



Problem 2 - A Monte Carlo Radiative Transfer code

1. In class, you were shown that the relation between ξ and μ/ϕ was

$$\mu_0 = 2\xi - 1$$

$$\phi_0 = 2\pi\xi$$

Show that this is true (submit your answer as a scan of a pen and paper solution). (2 marks)

start with $P(\mu, \phi) = \frac{1}{4\pi} d\mu d\phi$
which can be split into

$$P(\mu, \phi) d\mu d\phi = \frac{d\mu}{2} \frac{d\phi}{2\pi}$$

where $\mu = \cos\theta$ $d\mu = \sin\theta d\theta$

$$\therefore P(\mu) = \frac{1}{2} \quad \text{and} \quad P(\phi) = \frac{1}{2\pi}$$

The photon must come out at some μ interval -1 to 1 and at some ϕ in the interval 0 to 2π

Construct CDF

$$\psi(x_0) = \frac{\int_a^{x_0} P(x) dx}{\int_a^b P(x) dx} = \xi$$

For $\mu_0 =$ $P(\mu) = 1/2 \quad -1 \leq \mu \leq 1 \quad a = -1 \quad b = 1 \quad x_0 = \mu_0$

$$\xi = \frac{\int_{-1}^{\mu_0} 1/2 d\mu}{\int_{-1}^1 1/2 d\mu} = \frac{\frac{1}{2}(\mu_0 - (-1))}{\frac{1}{2}(1 - (-1))} = \frac{\mu_0 + 1}{2}$$

$$\xi = \frac{\mu_0 + 1}{2}$$

$$2\xi = \mu_0 + 1$$

$$\mu_0 = 2\xi - 1$$

For ϕ

$$p(\phi) = \frac{1}{2\pi} \quad 0 \leq \phi \leq 2\pi$$

$$a=0 \quad b=2\pi$$

$$x_0 = \phi_0$$

$$\xi = \frac{\int_0^{\phi_0} \frac{1}{2\pi} d\mu}{\int_0^{2\pi} \frac{1}{2\pi} d\mu} = \frac{\frac{1}{2}(\phi_0 - (0))}{\frac{1}{2}(2\pi - (0))} = \frac{\phi_0}{2} = \frac{\phi_0}{2\pi}$$

$$\xi = \frac{\phi_0}{2\pi} \Rightarrow \phi_0 = \xi 2\pi$$

2. Write a Monte Carlo code which simulates isotropic scattering of photons through a uniform slab of thickness $z_{\max} = 1$ and $\tau_{\max} = 10$. The photons should all be initialised at the origin with an initial angle of direction aligned with the z axis. They should then move a distance which is related to a randomly drawn optical depth, and then scatter into a random direction. The ending criterion for considering a photon should be when it leaves the atmosphere at $z = z_{\max}$ or back scatters out of the atmosphere. The final output of this program should be a histogram of the angles over which photons are emitted. Use this code to reproduce Chandrasekhar's result from 1960 for a plane atmosphere as shown below (and attached to this assignment as a dat file with columns of angle and intensity). Simulating roughly 100,000 photons should be enough to reproduce the result (depending on your bin size). Note that you'll need to convert from number of photons (flux) in a given bin to intensity. This can be done using

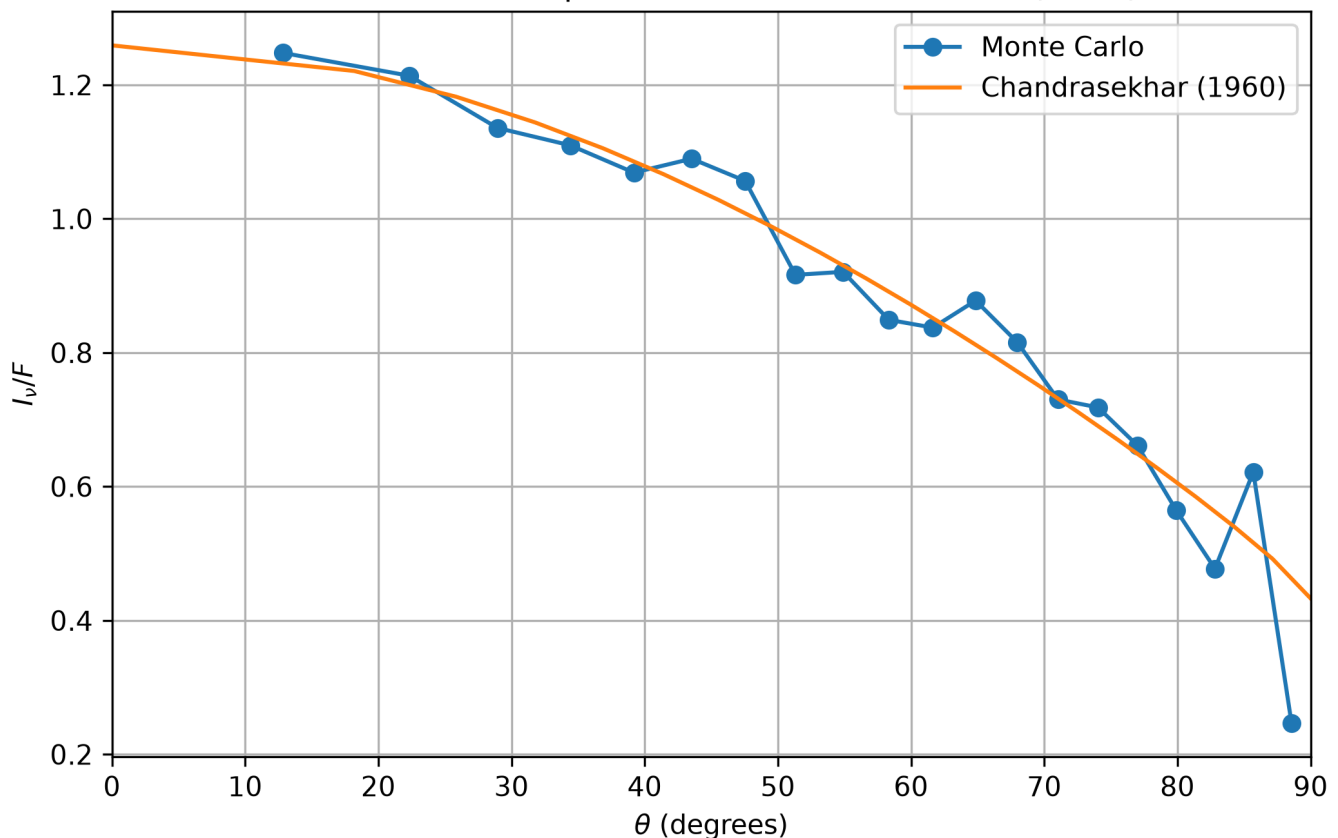
$$\frac{I_\nu}{F} = \frac{N_i N_\mu}{2N_0 \mu_i}$$

where N_i is the number of photons in bin i , N_μ is the total number of μ bins you split the data in to, N_0 is the total number of photons, and μ_i is the μ at the centre of the i th bin.

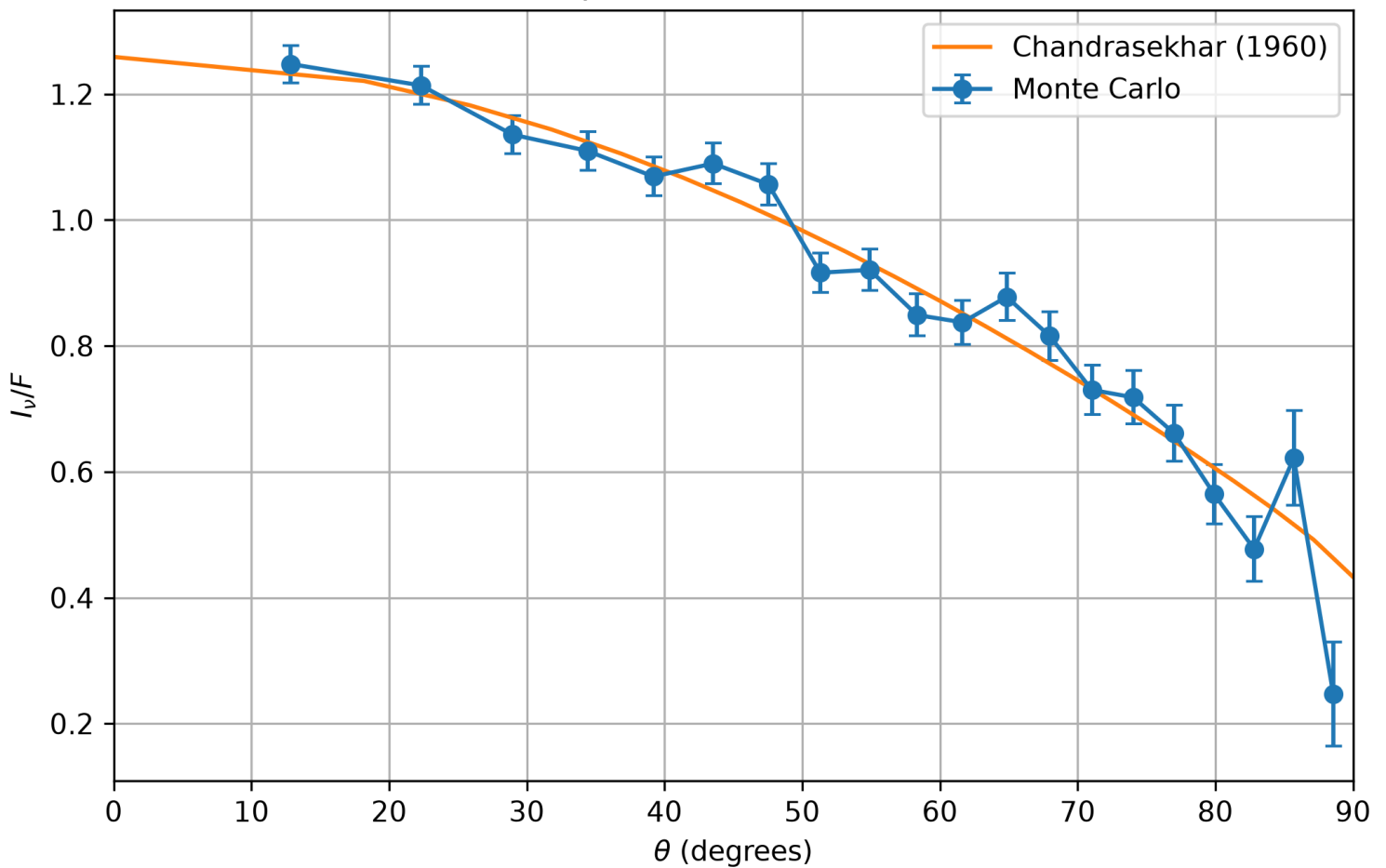
(10 marks)

3. Determine appropriate errors for each of your bins, and update your plot accordingly. Justify your choice of errors. (4 marks)

Monte Carlo Comparison with Chandrasekhar (1960)



Monte Carlo Comparison with Chandrasekhar (1960)



Justifying choice of errors:

uncertainty in number of photons in a bin

$$\sigma_{N_i} = \sqrt{N_i}$$

Propagate in Intensity:

$$\frac{I_v}{F} = \frac{N_i N_\mu}{2 N_o \mu_i}$$

$$\sigma I_{v/F} = \frac{\sqrt{N_i} N_\mu}{2 N_o \mu_i}$$

$\rightarrow$ N_i only thing not fixed

Relative uncertainty
higher in bins with fewer
photons

